

Integer Solutions of $y^2 + x^2y + xz^2 - 2 = 0$

Abstract

We give a complete elementary description of the integer solutions of

$$y^2 + x^2y + xz^2 - 2 = 0.$$

The solution set is split according to the sign of x . The case $x = 0$ is impossible. For $x > 0$, apart from two exceptional solutions, all solutions are obtained from a finite interval condition in the variable y . For $x < 0$, all solutions occur in two symmetric branches. We also exhibit an explicit Pell-type infinite subfamily, proving in particular that the full solution set is infinite.

1 Statement of the classification

We study the Diophantine equation

$$y^2 + x^2y + xz^2 - 2 = 0, \tag{1}$$

with $x, y, z \in \mathbb{Z}$. Equivalently,

$$y(y + x^2) + xz^2 = 2. \tag{2}$$

The following theorem is the complete classification.

Theorem 1.1. *The integer solutions of (1) are exactly the following.*

(i) *The two positive- x exceptional solutions*

$$(x, y, z) = (1, 1, 0), \quad (x, y, z) = (1, -2, 0).$$

(ii) *The positive- x middle family: for integers*

$$a \in \mathbb{Z}_{\geq 1}, \quad b \in \mathbb{Z}, \quad W \in \mathbb{Z}_{\geq 0},$$

satisfying

$$0 \leq b \leq a^2, \quad aW^2 = b(a^2 - b) + 2, \tag{3}$$

we obtain the solutions

$$(x, y, z) = (a, -b, \pm W). \tag{4}$$

Here, if $W = 0$, the two signs give the same solution.

(iii) *The negative- x family: for integers*

$$a \in \mathbb{Z}_{\geq 1}, \quad b \in \mathbb{Z}_{\geq 1}, \quad W \in \mathbb{Z}_{\geq 0},$$

satisfying

$$aW^2 = b(b + a^2) - 2, \tag{5}$$

we obtain the two symmetric branches

$$(x, y, z) = (-a, b, \pm W), \tag{6}$$

and

$$(x, y, z) = (-a, -a^2 - b, \pm W). \tag{7}$$

Again, if $W = 0$, the two signs give the same solution.

Conversely, every triple listed in (i), (ii), or (iii) is an integer solution of (1), and there are no other integer solutions.

The rest of the note is a proof of Theorem 1.1, followed by an explicit infinite subfamily.

2 Elementary preliminaries

Lemma 2.1 (No solution with $x = 0$). *There is no integer solution of (1) with $x = 0$.*

Proof. If $x = 0$, equation (1) becomes

$$y^2 - 2 = 0,$$

so $y^2 = 2$. No integer square is equal to 2. Hence no integer solution has $x = 0$. $\square$

Lemma 2.2 (The involution in y). *For fixed integers x and z , the expression*

$$y^2 + x^2y + xz^2 - 2$$

is invariant under the transformation

$$y \longmapsto -x^2 - y. \tag{8}$$

Consequently, if (x, y, z) is a solution of (1), then so is

$$(x, -x^2 - y, z).$$

Proof. Let $y' = -x^2 - y$. Then

$$y' + x^2 = -y.$$

Therefore

$$y'(y' + x^2) = (-x^2 - y)(-y) = y(y + x^2).$$

Since (1) is equivalent to

$$y(y + x^2) + xz^2 = 2,$$

the value of the left-hand side is unchanged when y is replaced by $y' = -x^2 - y$. This proves the claim. $\square$

Lemma 2.3 (Discriminant form). *Let $x, y, z \in \mathbb{Z}$. Then (x, y, z) satisfies (1) if and only if, with*

$$U = 2y + x^2, \tag{9}$$

one has

$$U^2 = x^4 - 4xz^2 + 8. \tag{10}$$

Equivalently, the solutions are exactly the triples obtained as

$$y = \frac{-x^2 + U}{2}, \tag{11}$$

where $x, z, U \in \mathbb{Z}$ satisfy (10). In this equivalent formulation, the integrality of y is automatic from (10).

Proof. Starting from (1), multiply by 4:

$$4y^2 + 4x^2y + 4xz^2 - 8 = 0.$$

Since

$$(2y + x^2)^2 = 4y^2 + 4x^2y + x^4,$$

we obtain

$$(2y + x^2)^2 = x^4 - 4xz^2 + 8.$$

Thus (10) holds with $U = 2y + x^2$.

Conversely, suppose $x, z, U \in \mathbb{Z}$ satisfy (10). We first check that $(-x^2 + U)/2$ is an integer. If x is odd, then

$$x^4 - 4xz^2 + 8 \equiv 1 \pmod{4},$$

so $U^2 \equiv 1 \pmod{4}$, and therefore U is odd. Thus U and x^2 have the same parity. If x is even, then

$$x^4 - 4xz^2 + 8 \equiv 0 \pmod{4},$$

so $U^2 \equiv 0 \pmod{4}$, and therefore U is even. Again U and x^2 have the same parity. Hence $y = (-x^2 + U)/2 \in \mathbb{Z}$.

For this integer y , we have $U = 2y + x^2$. Substituting this identity into (10) gives

$$(2y + x^2)^2 = x^4 - 4xz^2 + 8.$$

Expanding and cancelling x^4 gives

$$4y^2 + 4x^2y = -4xz^2 + 8.$$

Dividing by 4 gives

$$y^2 + x^2y + xz^2 - 2 = 0.$$

Thus (x, y, z) is a solution of (1). □

Remark 2.4. Lemma 2.3 is a compact complete description of the solution set: it says that x, z may be chosen exactly when $x^4 - 4xz^2 + 8$ is a square, and then y is forced up to the two choices $U = \pm\sqrt{x^4 - 4xz^2 + 8}$. The sign-split theorem below is more structural because it records the exact ranges forced by the sign of x .

3 The positive- x case

Lemma 3.1 (Positive x : the only outer solutions). *Let (x, y, z) be an integer solution of (1) with $x > 0$. Write $x = a$, where $a \in \mathbb{Z}_{\geq 1}$. If*

$$y \notin [-a^2, 0],$$

then

$$(a, y, z) = (1, 1, 0) \quad \text{or} \quad (a, y, z) = (1, -2, 0).$$

Proof. With $x = a > 0$, equation (2) becomes

$$y(y + a^2) + az^2 = 2. \tag{12}$$

Since $a > 0$ and $z^2 \geq 0$, we have

$$y(y + a^2) \leq 2. \tag{13}$$

Assume first that $y \geq 1$. Then

$$y(y + a^2) \geq 1(1 + a^2) = a^2 + 1.$$

Together with (13), this gives

$$a^2 + 1 \leq 2.$$

Since $a \in \mathbb{Z}_{\geq 1}$, this forces $a = 1$. Then the inequalities above force $y = 1$, because if $a = 1$ and $y \geq 2$, then

$$y(y + 1) \geq 2 \cdot 3 = 6 > 2.$$

Substituting $a = 1$ and $y = 1$ into (12) gives

$$1 \cdot 2 + z^2 = 2,$$

so $z^2 = 0$, hence $z = 0$. This gives $(a, y, z) = (1, 1, 0)$.

Assume now that $y \leq -a^2 - 1$. Then $y = -a^2 - t$ for some integer $t \geq 1$. Hence

$$y + a^2 = -t$$

and therefore

$$y(y + a^2) = (-a^2 - t)(-t) = t(a^2 + t) \geq a^2 + 1.$$

Again (13) gives $a^2 + 1 \leq 2$, so $a = 1$. With $a = 1$, if $t \geq 2$, then

$$t(1 + t) \geq 2 \cdot 3 = 6 > 2,$$

which is impossible. Hence $t = 1$, so $y = -2$. Substituting $a = 1$ and $y = -2$ into (12) gives

$$(-2)(-1) + z^2 = 2,$$

so $z^2 = 0$, hence $z = 0$. This gives $(a, y, z) = (1, -2, 0)$.

The two alternatives $y \geq 1$ and $y \leq -a^2 - 1$ are precisely the condition $y \notin [-a^2, 0]$. The lemma follows. $\square$

Lemma 3.2 (Positive x : the middle interval). *Let $a \in \mathbb{Z}_{\geq 1}$. The integer solutions of (1) with*

$$x = a, \quad -a^2 \leq y \leq 0$$

are exactly the triples

$$(x, y, z) = (a, -b, \pm W),$$

where $b, W \in \mathbb{Z}$ satisfy

$$0 \leq b \leq a^2, \quad W \geq 0, \quad aW^2 = b(a^2 - b) + 2.$$

Proof. Suppose first that (a, y, z) is a solution with $-a^2 \leq y \leq 0$. Put

$$b = -y, \quad W = |z|.$$

Then $0 \leq b \leq a^2$ and $W \in \mathbb{Z}_{\geq 0}$. Substituting $y = -b$ and $z^2 = W^2$ into (1) gives

$$b^2 - a^2b + aW^2 - 2 = 0.$$

Equivalently,

$$aW^2 = a^2b - b^2 + 2 = b(a^2 - b) + 2.$$

Thus every such solution gives parameters satisfying the stated condition.

Conversely, suppose $a \in \mathbb{Z}_{\geq 1}$, $b \in \mathbb{Z}$, and $W \in \mathbb{Z}_{\geq 0}$ satisfy

$$0 \leq b \leq a^2, \quad aW^2 = b(a^2 - b) + 2.$$

Set

$$x = a, \quad y = -b, \quad z = \pm W.$$

Then

$$\begin{aligned} y^2 + x^2y + xz^2 - 2 &= b^2 - a^2b + aW^2 - 2 \\ &= b^2 - a^2b + (b(a^2 - b) + 2) - 2 \\ &= 0. \end{aligned}$$

Hence each listed triple is indeed a solution. $\square$

Proposition 3.3 (Positive- x classification). *The integer solutions of (1) with $x > 0$ are exactly*

$$(1, 1, 0), \quad (1, -2, 0),$$

together with the family

$$(x, y, z) = (a, -b, \pm W)$$

where $a \in \mathbb{Z}_{\geq 1}$, $W \in \mathbb{Z}_{\geq 0}$, $0 \leq b \leq a^2$, and

$$aW^2 = b(a^2 - b) + 2.$$

Proof. Let (x, y, z) be a solution with $x > 0$, and write $x = a \in \mathbb{Z}_{\geq 1}$. If $y \notin [-a^2, 0]$, Lemma 3.1 gives exactly the two exceptional solutions. If $-a^2 \leq y \leq 0$, Lemma 3.2 gives exactly the stated family. The converse direction for the exceptional solutions is checked directly, and the converse direction for the family is proved in Lemma 3.2. Hence the proposition follows. $\square$

4 The negative- x case

Lemma 4.1 (Negative x : allowed ranges for y). *Let (x, y, z) be an integer solution of (1) with $x < 0$. Write $x = -a$, where $a \in \mathbb{Z}_{\geq 1}$. Then*

$$y > 0 \quad \text{or} \quad y < -a^2.$$

Equivalently, no solution with $x = -a$ has

$$-a^2 \leq y \leq 0.$$

Proof. With $x = -a$, equation (2) becomes

$$y(y + a^2) - az^2 = 2. \tag{14}$$

Thus

$$y(y + a^2) = 2 + az^2. \tag{15}$$

Because $a \geq 1$ and $z^2 \geq 0$, the right-hand side is at least 2. Hence

$$y(y + a^2) \geq 2 > 0. \tag{16}$$

If $-a^2 \leq y \leq 0$, then $y \leq 0$ and $y + a^2 \geq 0$, so

$$y(y + a^2) \leq 0,$$

contradicting (16). Therefore y cannot lie in $[-a^2, 0]$. Since y is an integer, this means $y > 0$ or $y < -a^2$. $\square$

Lemma 4.2 (Negative x : parametrization of both branches). *Let $a \in \mathbb{Z}_{\geq 1}$. The integer solutions of (1) with $x = -a$ are exactly the following triples. Choose $b \in \mathbb{Z}_{\geq 1}$ and $W \in \mathbb{Z}_{\geq 0}$ satisfying*

$$aW^2 = b(b + a^2) - 2.$$

Then take either

$$(x, y, z) = (-a, b, \pm W)$$

or

$$(x, y, z) = (-a, -a^2 - b, \pm W).$$

Proof. Let (x, y, z) be a solution with $x = -a$. By Lemma 4.1, either $y > 0$ or $y < -a^2$.

First suppose $y > 0$. Put

$$b = y, \quad W = |z|.$$

Then $b \in \mathbb{Z}_{\geq 1}$ and $W \in \mathbb{Z}_{\geq 0}$. Equation (14) gives

$$b(b + a^2) - aW^2 = 2,$$

that is,

$$aW^2 = b(b + a^2) - 2.$$

Thus the solution is of the form $(-a, b, \pm W)$ with the stated condition.

Now suppose $y < -a^2$. Put

$$b = -a^2 - y, \quad W = |z|.$$

Then $b \in \mathbb{Z}_{\geq 1}$, and $y = -a^2 - b$. Hence

$$y + a^2 = -b,$$

so

$$y(y + a^2) = (-a^2 - b)(-b) = b(b + a^2).$$

Using (14), we again obtain

$$b(b + a^2) - aW^2 = 2,$$

or equivalently

$$aW^2 = b(b + a^2) - 2.$$

Thus the solution is of the form $(-a, -a^2 - b, \pm W)$ with the stated condition.

Conversely, suppose $a \in \mathbb{Z}_{\geq 1}$, $b \in \mathbb{Z}_{\geq 1}$, and $W \in \mathbb{Z}_{\geq 0}$ satisfy

$$aW^2 = b(b + a^2) - 2.$$

For the first branch, set $(x, y, z) = (-a, b, \pm W)$. Then

$$\begin{aligned} y^2 + x^2y + xz^2 - 2 &= b^2 + a^2b - aW^2 - 2 \\ &= b(b + a^2) - aW^2 - 2 \\ &= 0. \end{aligned}$$

For the second branch, set $(x, y, z) = (-a, -a^2 - b, \pm W)$. Then

$$y + a^2 = -b,$$

so

$$y(y + a^2) = b(b + a^2).$$

Therefore

$$\begin{aligned} y^2 + x^2y + xz^2 - 2 &= y(y + a^2) - aW^2 - 2 \\ &= b(b + a^2) - aW^2 - 2 \\ &= 0. \end{aligned}$$

Hence every listed triple is a solution, and the lemma follows. $\square$

Proposition 4.3 (Negative- x classification). *The integer solutions of (1) with $x < 0$ are exactly the negative- x family in Theorem 1.1(iii).*

Proof. This is precisely Lemma 4.2. □

5 Proof of the complete classification

Proof of Theorem 1.1. By Lemma 2.1, no solution has $x = 0$. Hence every integer solution has either $x > 0$ or $x < 0$.

If $x > 0$, Proposition 3.3 gives exactly the two exceptional solutions and the positive- x middle family. If $x < 0$, Proposition 4.3 gives exactly the negative- x family. Each family has already been checked by direct substitution in the corresponding lemma. Therefore the three lists in Theorem 1.1 are necessary and sufficient, and there are no further integer solutions. □

6 An explicit infinite subfamily

The classification above already gives a complete description. We now show explicitly that the solution set is infinite.

Proposition 6.1 (A Pell-type infinite family). *Define integer sequences $(U_n)_{n \geq 0}$ and $(Z_n)_{n \geq 0}$ by*

$$U_0 = 573, \quad Z_0 = 60, \tag{17}$$

and

$$U_{n+1} = 33U_n + 272Z_n, \quad Z_{n+1} = 4U_n + 33Z_n. \tag{18}$$

Then, for every $n \geq 0$, the triples

$$\left(-17, \frac{U_n - 289}{2}, \pm Z_n \right) \tag{19}$$

and

$$\left(-17, \frac{-U_n - 289}{2}, \pm Z_n \right) \tag{20}$$

are integer solutions of (1). These solutions are infinitely many and pairwise distinct as n varies.

Proof. Set $x = -17$. Equation (1) becomes

$$y^2 + 289y - 17z^2 - 2 = 0. \tag{21}$$

Equivalently,

$$y^2 + 289y - 17z^2 = 2. \tag{22}$$

Multiplying by 4 and completing the square gives

$$(2y + 289)^2 - 68z^2 = 289^2 + 8.$$

Since

$$289^2 + 8 = 83521 + 8 = 83529,$$

we obtain the Pell-type equation

$$U^2 - 68z^2 = 83529, \quad U = 2y + 289. \tag{23}$$

The initial pair $(U_0, Z_0) = (573, 60)$ satisfies (23), because

$$573^2 - 68 \cdot 60^2 = 328329 - 244800 = 83529.$$

Also

$$33^2 - 68 \cdot 4^2 = 1089 - 1088 = 1.$$

We claim that every pair (U_n, Z_n) satisfies

$$U_n^2 - 68Z_n^2 = 83529. \quad (24)$$

Indeed, using (18),

$$\begin{aligned} U_{n+1}^2 - 68Z_{n+1}^2 &= (33U_n + 272Z_n)^2 - 68(4U_n + 33Z_n)^2 \\ &= (33^2 - 68 \cdot 4^2)U_n^2 \\ &\quad + (2 \cdot 33 \cdot 272 - 68 \cdot 2 \cdot 4 \cdot 33)U_nZ_n \\ &\quad + (272^2 - 68 \cdot 33^2)Z_n^2. \end{aligned}$$

The mixed coefficient is zero because $272 = 68 \cdot 4$. Moreover,

$$33^2 - 68 \cdot 4^2 = 1$$

and

$$272^2 - 68 \cdot 33^2 = 68^2 \cdot 4^2 - 68 \cdot 33^2 = 68(68 \cdot 16 - 33^2) = 68(1088 - 1089) = -68.$$

Therefore

$$U_{n+1}^2 - 68Z_{n+1}^2 = U_n^2 - 68Z_n^2.$$

Since the value is 83529 at $n = 0$, induction proves (24) for all $n \geq 0$.

Next we check integrality of the displayed y -coordinates. The integer $U_0 = 573$ is odd. If U_n is odd, then

$$U_{n+1} = 33U_n + 272Z_n$$

is odd, because $33U_n$ is odd and $272Z_n$ is even. Hence every U_n is odd. Since 289 is odd, both

$$\frac{U_n - 289}{2} \quad \text{and} \quad \frac{-U_n - 289}{2}$$

are integers.

By (24), the pair (U_n, Z_n) satisfies (23). Taking

$$U = U_n$$

gives

$$y = \frac{U_n - 289}{2},$$

and taking

$$U = -U_n$$

gives

$$y = \frac{-U_n - 289}{2}.$$

In both cases, (23) holds, and therefore the corresponding triples with $x = -17$ and $z = \pm Z_n$ satisfy (1) by the completed-square equivalence above, or by Lemma 2.3.

Finally, we show that these give infinitely many distinct solutions. Since $U_0 > 0$ and $Z_0 > 0$, the recursion gives $U_n > 0$ and $Z_n > 0$ for every n . Moreover,

$$U_{n+1} = 33U_n + 272Z_n > U_n.$$

Thus the sequence (U_n) is strictly increasing. Consequently the numbers

$$\frac{U_n - 289}{2}$$

are pairwise distinct, and hence the corresponding triples are pairwise distinct. Therefore (1) has infinitely many integer solutions. $\square$

Example 6.2. The first values of Proposition 6.1 give

$$(-17, 142, \pm 60), \quad (-17, -431, \pm 60),$$

then

$$(-17, 17470, \pm 4272), \quad (-17, -17759, \pm 4272),$$

and then

$$(-17, 1162126, \pm 281892), \quad (-17, -1162415, \pm 281892).$$

Each negative- y value is obtained from the corresponding positive- y value by the involution $y \mapsto -289 - y$ from Lemma 2.2.

7 Final conclusion

Combining Theorem 1.1 and Proposition 6.1, the equation

$$y^2 + x^2y + xz^2 - 2 = 0$$

has a complete elementary solution description and infinitely many integer solutions. The complete description is the sign-split family in Theorem 1.1; the infinite Pell-type subfamily in Proposition 6.1 is an explicit proper subfamily witnessing infinitude.